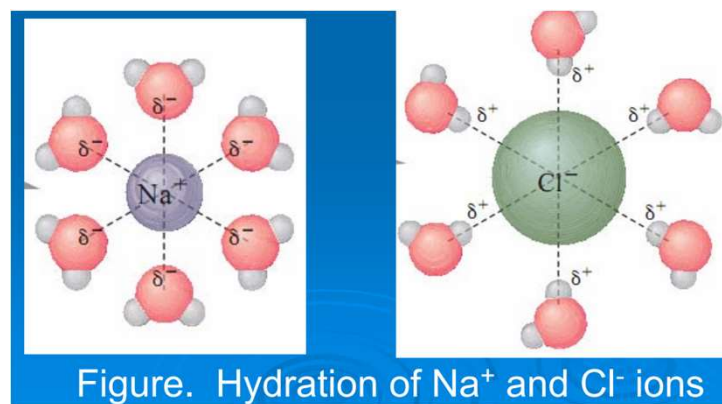
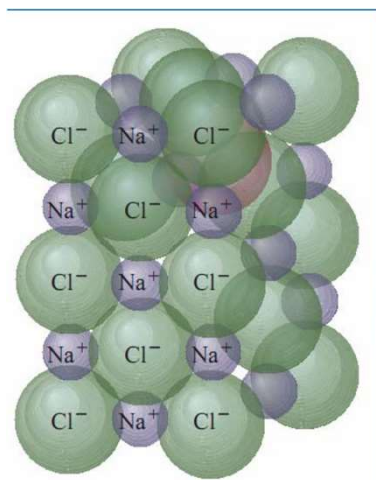


Roztoky

Elektrolyt – roztok, který vede elektrický proud

Elektrolytická disociace – S. Arrhenius 1887



ROZTOKY:

ELEKTROLYTICKÁ DISOCIACE

→ IONTY $\left\{ \begin{array}{l} \oplus \\ \ominus \end{array} \right.$

OHMŮV ZÁKON - TRÉNÍ

$$j = z N v$$

$$F = q E$$

$$F_{tr} = 6 \pi \eta r v \quad \left. \vphantom{F_{tr}} \right\} v = \frac{z}{6 \pi \eta}$$

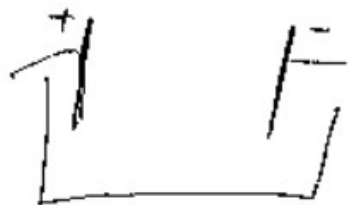
$$j = \frac{z^2 N}{6 \pi \eta} E$$

$$v = \text{konst.}$$

$$\sigma^+, \sigma^- \rightarrow \sigma = \sigma^+ + \sigma^-$$

ČASTO:
$$\sigma = \frac{z^2 N}{6 \pi \eta} \left(\frac{1}{z^+} + \frac{1}{z^-} \right)$$

FARADAYOVY ZÁKONY



PROUD PŘENÁŠEN ÍONTY

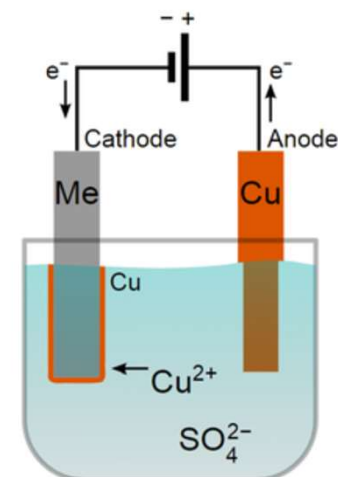
s VALENCÍ z ,
s HMOTNOSTÍ m

PROJDE NAŘÍSOJ $Q \rightarrow$ KOLIK ÍONTŮ? $N = \frac{Q}{ze}$

$\rightarrow$ JAKÁ HMOTNOST? $M = m N = \frac{m Q}{ze}$

$\Rightarrow \boxed{M = A Q}$ 1. FAR. ZÁKON

$A = \frac{m}{ze}$ ELEKTROCHEMICKÝ
EKVIVALENT



2 ROZTOKY A, B - STEJNÝ NÁBOJ Q

$$M_A = \frac{m_A Q}{z_A e}$$

$$M_B = \frac{m_B Q}{z_B e}$$

$$\frac{M_A}{M_B} = \frac{m_A/z_A}{m_B/z_B} = \frac{G_A}{G_B}$$

2. FAR. ZÁKON

CHEMICKÝ - GRAMEKVIVALENT

$$G = \frac{\text{HMOTNOST NOU LÁTKY}}{\text{VALENCE}} = \frac{m_{\text{NAVOB.}}}{z}$$

NÁBOJ NA 1 GRAMEKVIVALENT:

$$G = \frac{m_{\text{NAV}}}{z} = M = \frac{m Q_F}{z e} \Rightarrow F = Q_F = N_{\text{AVOB}} e = 9.64 \cdot 10^4 \text{ C/mol}$$

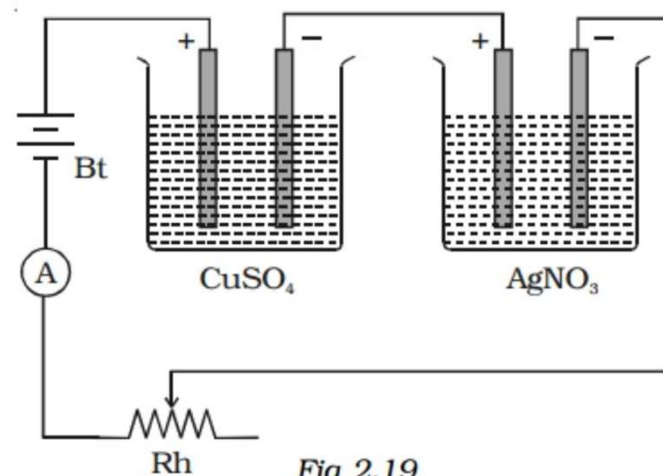


Fig 2.19

Verification of Faraday's second law